

ECON8037: Financial Economics

Week 7 Lecture - The Economics of Cryptocurrencies I: Blockchain, Mining, and Double-Spending (Chiu & Koepl, 2017)

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Why study the economics of cryptocurrencies?

Cryptocurrencies (Bitcoin, Ethereum, ...) are an attempt to support payments **without** a trusted third party.

- Critics: fraud, bubbles, vehicle for illegal activity.
- Advocates: cryptographically secure decentralised payments.
- Central banks: actively exploring blockchain-based digital currencies.

[Chiu and Koepl \(2017\)](#) ask: *how well can a cryptocurrency serve as a means of payment?*

- Build a general equilibrium monetary model with a blockchain;
- discipline it with Bitcoin transactions data;
- compute the welfare cost of using Bitcoin and an optimally designed alternative.

Roadmap for Lecture I

- ① Cryptocurrencies in one slide: blockchain, mining, PoW.
- ② The double-spending problem.
- ③ A partial-equilibrium model of mining and secret mining.
- ④ Lagos–Wright general equilibrium with a cryptocurrency.
- ⑤ Double-spending-proof equilibrium.

Lecture II picks up with optimal design, welfare, and applications.

Three key takeaways

- ① A cryptocurrency must overcome **double spending** by combining *costly mining* (proof-of-work) with *settlement lags* (confirmations).
- ② This creates a fundamental trade-off: settlement cannot be both **immediate** and **final**.
- ③ The model embeds these features in a Lagos–Wright environment so that the value of the currency, mining effort, and double-spending incentives are jointly determined.

Digital currency vs. cryptocurrency

A **digital token** is a string of bits. Digital strings can be copied costlessly $\Rightarrow$ the *double-spending problem*.

- **Centralised solution (e.g. PayPal):** a trusted third party maintains a ledger and updates it as people transact. Users must trust the intermediary.
- **Decentralised solution (e.g. Bitcoin):** a network of anonymous validators maintains and updates copies of the ledger. Trust is replaced by a *consensus protocol*.

The cryptocurrency must induce validators (i) to keep an accurate ledger and (ii) to deter users from double spending.

Blockchain as a record-keeping device

A **block** = a set of transactions conducted between users in a period. A

blockchain = a chain of blocks recording all past transactions.

- Each block is linked to the previous one (dynamically consistent ledger).
- Anyone can verify any user's balance by reading the chain.
- Different from traditional money, which only records the *current* distribution of balances.

Consensus rule: the **longest chain** is the trusted public record.

Mining and Proof-of-Work

Mining = competition to update the blockchain with a new block.

- A miner who first solves a computationally costly puzzle (*Proof-of-Work*, *PoW*) wins the right to append a new block.
- Winning probability is proportional to the miner's fraction of computational power.
- Winners receive a reward R , financed by (i) newly created coins (seigniorage) and (ii) transaction fees.

Why is mining costly? To make rewriting history expensive. Older transactions are more trustworthy because revising them requires re-doing the PoW for every subsequent block.

The bootstrapping problem

A cryptocurrency must jointly support three things:

- **Security of the blockchain** – so merchants accept the currency;
- **Value of the currency** – so mining rewards are valuable;
- **Mining activity** – so double spending is costly.

Each side reinforces the others. The model must determine all three jointly in equilibrium.

The double-spending attack: forward looking

Even with a blockchain, a *forward-looking* attack is possible:

- ① Buyer sends payment d to seller; instruction is broadcast to miners.
- ② Buyer secretly mines an *alternative* block in which the payment does not occur.
- ③ Once seller delivers the goods, buyer releases the secret chain.
- ④ If the secret chain is longer, the consensus protocol adopts it and the original payment is wiped out.

Outcome: buyer keeps both the goods and the balances; seller is left empty handed.

Confirmation lags as a defence

Defence: seller waits for N *confirmations* (i.e. N blocks added on top of the block containing the payment) before delivering goods.

- To double spend, buyer must mine $N + 1$ **blocks in a row** faster than the rest of the network.
- Longer $N \Rightarrow$ harder to attack, but slower settlement.

Trust = (security of blockchain) $\times$ (mining ecosystem) $\times$ (value of currency).

Partial equilibrium: one transaction period

Within a single period there are $\bar{N} + 1$ subperiods. Buyer holds real balances z ; bargains with a seller over (x, d, N) :

- x = quantity of goods,
- $d \leq z$ = payment in real balances,
- $N \in \{0, \dots, \bar{N}\}$ = confirmation lag.

Buyer's preferences (with confirmation lag N):

$$\delta^N u(x), \quad \delta \in (0, 1).$$

Across periods agents discount by $\beta = \delta^{\bar{N}+1}$.

The mining game

M miners each invest computing power $q(i)$ measured in real balances. A particular miner wins block n with probability

$$\rho(i) = \frac{q(i)}{\sum_{m=1}^M q(m)}. \quad (3)$$

Each subperiod, each miner solves

$$\max_{q(i)} \rho_i \beta R - q(i). \quad (4)$$

Symmetric Nash equilibrium $q(m) = Q$ for all m :

$$Q = \frac{M-1}{M^2} \beta R. \quad (6)$$

Free entry of miners

Total computing cost per subperiod:

$$MQ = \frac{M-1}{M} \beta R. \quad (7)$$

Take $M \rightarrow \infty$ (competitive mining):

Lemma (Rent dissipation)

As $M \rightarrow \infty$, miners earn zero profits and aggregate computing power dissipates all rewards:

$$MQ = \beta R.$$

Mining is a pure rent-seeking activity: every dollar of reward is matched by a dollar of cost.

Secret mining and double-spending payoffs

To double spend successfully, the buyer must win the mining game in $N + 1$ consecutive subperiods. Conditional on N prior successes, his subperiod- N payoff from investing q_N is

$$\rho(q_N) \beta [d + (N + 1)R] - q_N. \quad (9)$$

Define

$$\Delta \equiv \frac{d}{R} + (N + 1). \quad (11)$$

As $M \rightarrow \infty$:

$$\hat{q}_N(d, N) = \beta R [\sqrt{\Delta} - 1], \quad \rho(\hat{q}_N) = \frac{\sqrt{\Delta} - 1}{\sqrt{\Delta}}. \quad (12,13)$$

Backward induction

Define recursively the expected double-spending payoff in subperiod n :

$$D_n(d, N) = \max_{q_n} \rho(q_n) D_{n+1}(d, N) - q_n, \quad D_N(d, N) = \beta R(\sqrt{\Delta} - 1)^2.$$

Lemma (Closed form)

For $s = 0, \dots, N$,

$$D_{N-s}(d, N) = \beta R[\sqrt{\Delta} - (s + 1)]^2, \quad \hat{q}_{N-s} = \beta R[\sqrt{\Delta} - (s + 1)].$$

Investment is increasing in n : if it's optimal to engage in secret mining today, it's optimal to continue tomorrow conditional on success.

The no-double-spending (DS) constraint

Double spending is unattractive whenever

$$\beta R[\sqrt{\Delta} - (N + 1)] < 0,$$

so $\hat{q}_0 = 0$ and $D_0 = 0$. This yields the central inequality.

Proposition (DS-proof contract)

As $M \rightarrow \infty$, a contract (d, N) is double-spending proof iff

$$d < R(N + 1)N. \quad (20)$$

Intuition:

- Larger payment $d \Rightarrow$ stronger incentive to attack.
- Larger reward R or longer lag $N \Rightarrow$ harder to attack.

Probability of successful double spending

More generally, whenever (20) fails, the unconditional probability of a successful attack is

$$P(d, N) = \frac{\sqrt{\Delta} - (N + 1)}{\sqrt{\Delta}}, \quad \Delta = \frac{d}{R} + (N + 1).$$

Settlement concepts.

- Settlement is *immediate* if $N = 0$, *delayed* if $N > 0$;
- Settlement is *final* if $P(d, N) = 0$, *probabilistic* if $P > 0$.

Fundamental impossibility

Theorem (Chiu–Koepl)

For any cryptocurrency based on a PoW protocol, settlement cannot be both immediate and final.

Why? At $N = 0$ the DS constraint $d < R(N + 1)N$ fails for any $d > 0$. Final settlement requires $N \geq 1$. This is a built-in cost of decentralised consensus: trust must be earned by waiting.

Embedding the model in Lagos–Wright

Chiu and Koepl (2017) embed the partial equilibrium model in a standard Lagos–Wright (2005) environment.

- Time $t = 0, 1, \dots$; continuum of buyers B , sellers $S = B\sigma$, $\sigma \in (0, 1)$.
- Each period: **day market** (centralised, general good h) then **night market** ($\bar{N} + 1$ subperiods).
- In the first subperiod, buyers match with sellers w.p. σ .
- Buyer's preference shock $\varepsilon \sim F_\varepsilon$ at start of period.
- Utility from night-market consumption: $\delta^N \varepsilon u(x)$.

Cryptocurrency in general equilibrium

The cryptocurrency m is a digital record of balances.

- Day market: perfect monitoring, no double spending.
- Night market: anonymous; cryptocurrency is the only means of payment.
- PoW protocol updates the chain each subperiod.

The mining reward is endogenous and pinned down by design parameters:

$$R = \frac{Z(\mu - 1) + D\tau}{\bar{N} + 1}, \quad (23)$$

where $\mu \geq 1$ is the gross money growth rate, $\tau \geq 0$ the transaction fee, $Z = BE(z)$ aggregate balances, $D = B\sigma E(d)$ aggregate transfers.

Day market value function

Let $z = \phi m$ be real balances. The day-market problem of a buyer is

$$w(z) = \max_{z', h} -h + v(z', \varepsilon) \quad \text{s.t.} \quad z + h \geq z' \geq 0. \quad (24, 25)$$

Linearity of utility in h implies

$$w(z) = z + w(0), \quad \mathbb{E}[w(z, \varepsilon)] = z + W. \quad (27, 28)$$

Standard Lagos–Wright result: balances enter linearly into the value function.

Night market and buyer's problem

A matched buyer holding balances z makes a TIOLI offer (x, d, N) to the seller.
Buyer's night-market value:

$$v(z; \varepsilon) = \frac{\beta}{\mu} z + \sigma[\delta^N \varepsilon u(x) + D_0(d, N) - \frac{\beta}{\mu} d] + \beta W. \quad (29)$$

Seller's value:

$$v(z; 0) = \frac{\beta}{\mu} d(1 - \tau)(1 - P(d, N)) - x + \frac{\beta}{\mu} z + \beta W. \quad (30)$$

Cash-in-advance: with $\mu \geq 1 > \beta$, sellers carry no balances and buyers carry exactly $z' = d$.

Buyer's optimal contract

Buyer solves

$$\max_{(x,d,N)} -d + (1 - \sigma) \frac{\beta}{\mu} d + \sigma [\delta^N \varepsilon u(x) + D_0(d, N)]$$

subject to the seller's participation constraint:

$$x = \frac{\beta}{\mu} d (1 - \tau) (1 - P(d, N)). \quad (33)$$

Two possible regimes:

- DS-proof: $d \leq R(N + 1)N$, so $P = D_0 = 0$.
- DS-attack: $P > 0$, $D_0 > 0$ – attractive only when balances are very costly to carry.

DS-proof equilibrium

Aggregate transfers and money demand:

$$D = B\sigma \int_0^\infty z^*(\varepsilon; R) dF_\varepsilon(\varepsilon), \quad Z = B \int_0^\infty z^*(\varepsilon; R) dF_\varepsilon(\varepsilon). \quad (35,36)$$

Definition (DS-proof cryptocurrency equilibrium)

Given (μ, τ) : offers $(x(\varepsilon), d(\varepsilon), N(\varepsilon))$, money demand $z(\varepsilon) > 0$ and mining choice q such that

- ① the buyer's contract maximises utility;
- ② mining is a Nash equilibrium in every subperiod;
- ③ the contract satisfies $d \leq R(N + 1)N$;
- ④ the day market for balances clears.

Existence

Proposition (Existence)

If B is sufficiently high, a DS-proof cryptocurrency equilibrium exists.

The key insight:

- Mining reward R is financed by **aggregate** transactions (scales with B).
- Incentive to double spend depends only on **individual** transaction size d .
- So large $B \Rightarrow$ large $R \Rightarrow$ DS constraint $d < R(N + 1)N$ is easy to satisfy.

Cryptocurrencies are essentially *scale-dependent*: they work better when the user base is large.

Summary of Lecture I

- ① Cryptocurrency = digital token + blockchain + PoW + consensus rule.
- ② Double-spending attacks deterred by (i) costly mining and (ii) confirmation lags.
- ③ Closed-form attack technology gives a tractable DS constraint $d < R(N+1)N$.
- ④ Free-entry of miners $\Rightarrow$ all rewards dissipated as mining cost ($MQ = \beta R$).
- ⑤ Lagos–Wright embedding determines R jointly with the value of the currency and individual contracts.
- ⑥ Settlement is never simultaneously immediate *and* final.

Next lecture: optimal design (μ, τ) , calibration to Bitcoin, welfare, applications.

References I

Chiu, J. and T. V. Koepl (2017). The Economics of Cryptocurrencies – Bitcoin and Beyond. *Bank of Canada Staff Working Paper / Queen's University*.

Lagos, R. and R. Wright (2005). A Unified Framework for Monetary Theory and Policy Analysis. *Journal of Political Economy* 113(3), 463–484.